



University of Engineering & Technology Peshawar

Computer Programming (C++) Lab

Lab 06

STUDENT NAME	Muhammad Yaseen
REGISTRATION NO:	BFNWELE0670
COUSRE:	EE-170L COMPUTER PROGRAMMING LAB
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LAB TITLE:	ONE DIMENSIONAL ARRAY IN C++

SUBMITTED TO ENGR RIZWAN ULLAH

Learning Objectives:

The objective of this lab is to understand and apply the for loop in C++. Students will learn how to:

- Use a for loop to repeat a block of code a fixed number of times.
- Use nested for loops to work with patterns and tables.
- Calculate sums using for loops.
- Print patterns using nested for loops.

Introduction:

A for loop is a repetition control structure that allows a programmer to write a loop that executes a specific number of times. It is used when a variable runs from a starting point to an ending point with a constant increment or decrement.

Syntax of for loop:

```
for (initialization; condition; update)
{
    // statements
}
```

A nested for loop is a for loop placed inside another for loop. It is commonly used for printing patterns or generating tables.

Task 1: for Loop – Display Numbers 1 to 10

Write a C++ program that uses a for loop to display numbers from 1 to 10, each on a separate line.

Code:

```
#include <iostream>
using namespace std;
int main()
{
    for (int i = 1; i <= 10; i++)
    {
        cout << i << endl;
    }
    return 0;
}
```

Output:

```
1
2
3
4
5
6
7
8
9
10
```

Task 2: Nested for Loop – Multiplication Table

Write a C++ program that generates a multiplication table from 1 to 10 using nested for loops.

Code:

```
#include <iostream>
```

```
#include <iomanip>
using namespace std;
int main()
{
    for (int i = 1; i <= 10; i++)
    {
        for (int j = 1; j <= 10; j++)
        {
            cout << setw(4) << i * j;
        }
        cout << endl;
    }
    return 0;
}
```

Output (first 3 rows shown):

```
1  2  3  4  5  6  7  8  9 10
2  4  6  8 10 12 14 16 18 20
3  6  9 12 15 18 21 24 27 30
... (continues up to row 10)
```

Task 3: Sum Using for Loop

Write a C++ program that prompts the user to enter a positive integer and uses a for loop to calculate the sum of all numbers from 1 to that integer.

Code:

```
#include <iostream>
using namespace std;
int main()
{
    int n, sum = 0;
    cout << "Enter a positive integer: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
    {
        sum = sum + i;
    }
    cout << "Sum from 1 to " << n << " = " << sum << endl;
    return 0;
}
```

Output:

```
Enter a positive integer: 5
Sum from 1 to 5 = 15
```

Task 4: Nested Pattern Printing

Write a C++ program that uses nested for loops to print a triangle pattern of asterisks. The number of rows is determined by user input.

Code:

```
#include <iostream>
using namespace std;
int main()
{
    int rows;
    cout << "Enter number of rows: ";
    cin >> rows;
    for (int i = 1; i <= rows; i++)
    {
        for (int j = 1; j <= i; j++)
        {
```

```
        cout << "* ";  
    }  
    cout << endl;  
}  
return 0;  
}
```

Output (for rows = 5):

```
*  
* *  
* * *  
* * * *  
* * * * *
```

Conclusion:

In this lab, we successfully learned and applied the for loop in C++. We used it to display numbers, calculate sums, generate a multiplication table, and print triangle patterns. The nested for loop proved especially useful for working with rows and columns. These exercises helped us understand how loops control program flow and reduce repetitive code.

BFNWLE0670